POSITIVE-DEFINITENESS OF NTK FOR GENERAL NEURAL NETWORK ARCHITECTURES

Yash Jakhmola
yash.jakhmola01@gmail.com

ABSTRACT

A key challenge in modern deep learning theory is to explain the remarkable success of gradient-based optimization methods when training large-scale, complex deep neural networks. It is well-known that linear convergence of such methods can be proved once it is shown that the minimum eigenvalue of the NTK is bounded away from zero, ie. $\lambda_{\text{min}} \geq c$ for some constant $c > 0$. Here, we state an interesting finding that the lowest eigenvalue remains positive (this is not enough for linear convergence though) throughout gradient flow training of any (mildly) over-parametrized neural network with polynomial layers and piecewise nonzero polynomial activation functions almost surely (with respect to initialization and input distributions).

1 Setup and Preliminaries

Consider a continuous function $f : \mathbb{R}^N \times \mathbb{R}^P \to \mathbb{R}^M$ defined as follows for some $N, P, M \in \mathbb{N}$:

$$f(X, \theta) := g_L(\sigma(\ldots g_2(\sigma(g_1(X, \theta)))\ldots)) \tag{1}$$

for all inputs $X \in \mathbb{R}^N$ and parametrizations $\theta \in \mathbb{R}^P$, where $g_i : \mathbb{R}^{N_{i-1}} \times \mathbb{R}^P \to \mathbb{R}^{N_i}$ are polynomials (in each component) for all $i \in [L]$ with $N_0 = N, N_L = M$ and $N_1, \ldots, N_{L-1} \in \mathbb{N}$ for some $L \in \mathbb{N}$. $\sigma : \mathbb{R} \to \mathbb{R}$ is a continuous, piecewise polynomial (with finitely many closed pieces) which is zero at finitely many values only. Note that σ applied to a vector means that it is applied component-wise.

We intend to find that parametrization of f that minimizes its error over a training dataset $(X_1, y_1), \ldots, (X_n, y_n) \in \mathbb{R}^N \times \mathbb{R}^M$, where $L : \mathbb{R}^P \to [0, \infty)$ is the error, defined by $L(\theta) := \frac{1}{2}\sum_{j=1}^{n} ||f(X_j, \theta) - y_j||^2$ for all $\theta \in \mathbb{R}^P$.

To do so, we utilize gradient flow. Start from some initial parameters θ_0 and input data $X_1, \ldots, X_n$ sampled from some arbitrary absolutely continuous probability distributions (with respect to the Lebesgue measure). Then, for some $T > 0$, solve the following initial-value problem till $t = T$ to get the final parametrization $\theta(T)$.

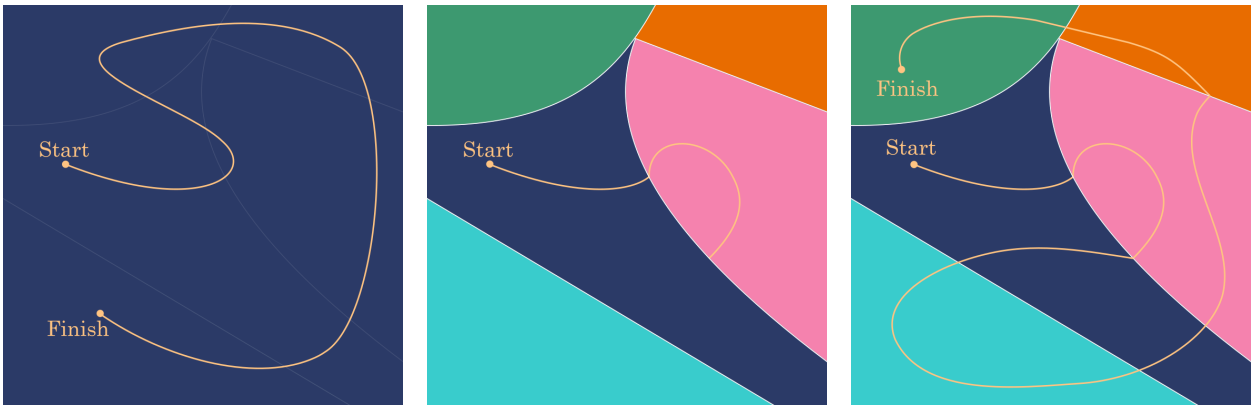
$$\begin{aligned} \frac{\mathrm{d}\theta(t)}{\mathrm{d}t} &= -\frac{\partial L(\theta(t))}{\partial \theta(t)} \\ \theta(0) &= \theta_0 \end{aligned} \tag{2}$$

We shall often write $\nabla L(\theta(t))$ in place of $\frac{\partial L(\theta(t))}{\partial \theta(t)}$. Note that the f need not be differentiable with respect to θ everywhere, but it is differentiable almost everywhere, due to the definition of σ. Note that this is will not be an issue, since the ODE theorems are applied piecewise and the training dynamics are not affected by almost everywhere differentiability.

The following lemma will prove that this ODE has a unique continuous and piecewise differentiable solution that can be extended till any $T > 0$.

Lemma 1.1 (Existence and invertibility of GF solution curve)**.** *The GF initial-value problem (2) has a unique, continuous and piecewise differentiable solution that can be extended till any $T > 0$, for almost all initializations and input data. Moreover, the solution flow is a diffeomorphism over the parameter space minus a measure zero set.*

Proof. Note that we assume $\nabla L(\theta(0)) \neq 0$ to exclude the trivial case of constant solution. Since GF cannot reach such a critical point, we need not worry about this case for the inverse solution. Moreover, we assume the initialization does



(a) Solution curve of the ODE in the initialization (blue) piece till finish ($t = T$). We cut off this solution at the point where it exits the initialization piece.

(b) We continue the solution by solving the ODE in the next (pink) piece and continue till we exit it.

(c) We keep on finding solutions for every piece and join them together to form the final solution. The curve will not hit an intersection of more than two pieces for almost all initializations.

Figure 1: Constructing the solution curve through the parameter space.

not occur on a boundary, in order to apply Picard's theorem. Since boundaries and the set $\{\theta \in \mathbb{R}^P : \nabla L(\theta) = 0\}$ have zero measure (since $\nabla L(\theta)$ is a piecewise nonzero polynomial for almost all input datasets), the following arguments hold for almost all intializations amd input data.

Consider $-\nabla L(\theta)$ in its initialization piece. Then, it is a polynomial in θ, and thus is Lipschitz continuous in any closed ball around the initialization θ_0. By Picard's existence and uniqueness theorem [Hartman, 1964] the GF equation has a unique solution in a neighbourhood of $t = 0$. Since there is no finite time blowup of the solution (see theorem A.3), it can be extended beyond a neighbourhood of $t = 0$ [Hartman, 1964], either to all $t > 0$ or till it hits the boundary of its piece.

If this solution hits the boundary of the piece containing the initial condition $\theta(0)$, we can construct solution for the same ODE in the next piece, with initial condition being defined as the end point of the previous solution curve. Note that the curve will reach an intersection of more than two pieces for a zero measure set of initializations (see theorem A.1). Consequently, we do not need to worry about having to choose between two or more pieces for the solution curve to continue into.

Finally, these solutions can be joined together continuously (due to no finite time blowup, see theorem A.3) until we reach $t = T$. Define this piecewise differentiable, but continuous everywhere solution of the GF equation by $\Phi_t(\theta_0) := \theta(t)$ for all $t \geq 0$.

Now, note that the Φ_t is a piecewise diffeomorphism over the initial parameter space minus a measure zero set (ie. excluding critical points, boundary points and points with image/preimage curve that reaches an intersection of more than two pieces) since $-\nabla L(\theta(t))$ is piecewise continuously differentiable with respect to initial conditions [Hartman, 1964], all piecewise solutions are invertible as they are solutions of an autonomous system of ODEs, which all exist and can be joined uniquely by the above arguments, and the inverse solution $\Phi_t^{-1} = \Phi_{-t_0}$ is also piecewise differentiable (again by previous arguments). $\square$

2 Main Result

The NTK, denoted as $G(t)$ here, is the gram matrix at time t of training defined as follows.

$$G(t) := \left(\left(\frac{\partial F_t}{\partial \theta(t)} \right)_p^T \left(\frac{\partial F_t}{\partial \theta(t)} \right)_q \right)_{p,q=1}^{nM} \tag{3}$$

where $\frac{\partial F_t}{\partial \theta(t)} = \left(\frac{\partial (f(X_1, \theta(t)))_1}{\partial \theta(t)}, \ldots, \frac{\partial (f(X_1, \theta(t)))_M}{\partial \theta(t)}, \ldots, \frac{\partial (f(X_n, \theta(t)))_1}{\partial \theta(t)}, \ldots, \frac{\partial (f(X_n, \theta(t)))_M}{\partial \theta(t)} \right)$ is a $P \times nM$ matrix and $P \in \mathbb{N}$ is the number of trainable parameters.

$G(t)$ is a nonegative definite matrix, since it is a gram matrix. It can be shown that if its lowest eigenvalue is larger than a positive constant, then we get linear convergence of the optimizer Du et al. [2019b,a], Allen-Zhu et al. [2019b,a]. The lowest value has been bounded away from zero for specific architectures Nguyen et al. [2021].

Theorem 2.1 (Positive definiteness of NTK)**.** *$G(t)$ is a positive definite matrix almost surely (with respect to initialization and input data) for all $t \geq 0$, for networks of form (1) with $P \geq nM$.*

Proof. We know that $G(t)$ is positive definite iff $\left\{ \left(\frac{\partial F_t}{\partial \theta(t)} \right)_1, \ldots, \left(\frac{\partial F_t}{\partial \theta(t)} \right)_{nM} \right\} \subset \mathbb{R}^P$ is linearly independent. Note that this set is always linearly dependent if $nM > P$. Thus, we shall assume that $nM \leq P$. We therefore need the function to be overparametrized with respect to the training sample size.

If this set is linearly dependent at $t = 0$, then all sub-square matrices of $\left(\left(\frac{\partial F_0}{\partial \theta(0)} \right)_1 \cdots \left(\frac{\partial F_0}{\partial \theta(0)} \right)_{nM} \right) \in \mathbb{R}^{P \times nM}$ will be singular. Note that the determinants of these matrices are piecewise polynomials in the parameters and the dataset and σ is zero only at finitely many points, since it is a piecewise nonzero polynomial. Thus, the determinants are nonzero for Lebesgue-almost all values of the parameters $\theta(0)_1, \ldots, \theta(0)_P$ and the datasets $X_1, \ldots, X_n$ [Caron, 2005]. Let θ_{degen} denote the set of those parameters for which the determinants are zero, union the set of degenerate parameters from the proof of lemma 1.1. Due to the continuous distribution of initial parameters and input data, we conclude that $\mathbb{P}(\theta(0) \in \theta_{\text{degen}}) = 0$.

We shall now show that $\mathbb{P}(\theta(t) \in \theta_{\text{degen}}) = 0$ for all $t > 0$. From lemma 1.1, we know that the GF solution flow is a piecewise diffeomorphism over the initial conditions space minus a zero measure set. Thus, it maps zero probability sets to zero probability sets (see theorem A.4). Now, we have shown that with respect to initialization and input data, $\mathbb{P}(\theta_{\text{degen}}) = 0$. Thus, from above arguments, $\mathbb{P}(\Phi_{-t}(\theta_{\text{degen}})) = 0$, which implies that $\mathbb{P}(\theta(0) |\ \Phi_t(\theta(0)) \in \theta_{\text{degen}}) = \mathbb{P}(\theta(t) \in \theta_{\text{degen}}) = 0$. □

Remark. An analogous proof holds for matrix-valued inputs/outputs by flattening the input/output matrix into a vector.

References

Zeyuan Allen-Zhu, Yuanzhi Li, and Yingyu Liang. Learning and generalization in overparameterized neural networks, going beyond two layers. *Advances in neural information processing systems*, 32, 2019a.

Zeyuan Allen-Zhu, Yuanzhi Li, and Zhao Song. A convergence theory for deep learning via over-parameterization. In *International conference on machine learning*, pages 242–252. PMLR, 2019b.

Richard Caron. The zero set of a polynomial. *10.13140/RG.2.1.4432.8169*, 05 2005.

E W Cheney. *Introduction to Approximation Theory*. AMS Chelsea Publishing. American Mathematical Society, 2 edition, 1998.

Simon Du, Jason Lee, Haochuan Li, Liwei Wang, and Xiyu Zhai. Gradient descent finds global minima of deep neural networks. In *International conference on machine learning*, pages 1675–1685. PMLR, 2019a.

Simon S Du, Xiyu Zhai, Barnabás Poczós, and Aarti Singh. Gradient descent provably optimizes over-parameterized neural networks. In *International Conference on Learning Representations*, 2019b.

Alfred Gray. *Tubes*. Progress in Mathematics. Birkhauser Verlag AG, 2 edition, 2003.

Philip Hartman. *Ordinary Differential Equations*. John Wiley & Sons, Inc., 1964.

Quynh Nguyen, Marco Mondelli, and Guido F Montufar. Tight bounds on the smallest eigenvalue of the neural tangent kernel for deep relu networks. In *International Conference on Machine Learning*, pages 8119–8129. PMLR, 2021.

Walter Rudin. *Real and Complex Analysis*. McGraw-Hill Education, 3rd edition, 2006.

A Auxillary Results

Theorem A.1. *The solution curve of the GF equation 2 reaches a subspace S of codimension $k \geq 2$ for almost no initializations.*

Proof. Define the set of bad initializations as

$$A := \{\theta_0 \in \mathbb{R}^P :\ \exists t \geq 0 \text{ such that } \Phi_t(\theta_0) \in S\} \tag{4}$$

We shall show that $\lambda^P(A) = 0$. Define the earliest time that the curve hits S, for $\theta \in A$ by

$$t_0(\theta) := \inf\{t > 0 : \ \Phi_t(\theta) \in S\} \tag{5}$$

Define the r-neighbourhood around S for any $r > 0$ by

$$N_r(S) := \{\theta \in \mathbb{R}^P : \ ||x - S|| < r\} \tag{6}$$

Since the flow is Lipschitz continuous in t, $||\Phi_t(\theta_0) - \Phi_q(\theta_0)|| \leq L|t - q|$ for all $t, q \geq 0$. Choose a finite rational grid Q_r with spacing $\tau \leq r/L$. Note that we cannot choose a finer spacing as that will allow $\Phi_q(\theta) \notin N_r(S)$, which will cause a contradiction later on in the proof.

Thus, for all $t \geq 0$, we can find $q \in Q_r$ such that $||\Phi_t(\theta_0) - \Phi_q(\theta_0)|| \leq r$.

Furthermore, since Φ_q is C^1 almost everywhere, $\lambda^P(\Phi_q^{-1}(N_r(S))) = \int_{N_r(S)} |\det D\Phi_q^{-1}(y)|dy \leq J\lambda^P(N_r(S))$ for some constant $J > 0$. So, putting all of the above results together,

$$\begin{aligned} \lambda^P(A) &\leq \sum_{q \in Q_r} \lambda^P(\Phi_q^{-1}(N_r(S))) && (7) \\ &\leq |Q_r| J \lambda^P(N_r(S)) && (8) \\ &= \mathcal{O}(r^{k-1}) && (9) \end{aligned}$$

where last line follows from the fact that volume of $N_r(S)$ is $\mathcal{O}(r^k)$ as $r \to 0$, which follows from Weyl's tube formula (see Gray [2003]). Finally, taking $r \to 0$, we get our conclusion. □

Theorem A.2 (Jackson's inequality)**.** *Let $k \in \mathbb{N}$ and $a < b$. Any k-times continuously differentiable function $f : [a, b] \to \mathbb{R}$ can be uniformly approximated by a sequence of polynomials (p_n) such that*

$$||f - p_n||_\infty \leq \frac{\pi}{2} \frac{|f^{(k)}|}{(n+1)^k} \tag{10}$$

Proof. See Cheney [1998]. □

Theorem A.3. *Solution of the GF equation 2 does not blow up in finite time, ie. $||\theta(s)|| < \infty$ for all $s \in (0, \infty)$.*

Proof. Note that

$$\frac{\mathrm{d}L(\theta(t))}{\mathrm{d}t} = \nabla L(\theta(t))^T \left(\frac{\mathrm{d}\theta(t)}{\mathrm{d}t} \right) = -||\nabla L(\theta(t))||^2 \tag{11}$$

Thus,

$$\int_0^s ||\nabla L(\theta(t))||^2 dt = -\int_0^s \frac{\mathrm{d}L(\theta(t))}{\mathrm{d}t} dt = L(\theta(0)) - L(\theta(s)) \leq L(\theta(0)) < \infty \tag{12}$$

Now,

$$||\theta(s) - \theta(0)|| = \left|\left| \int_0^s \frac{\mathrm{d}\theta(t)}{\mathrm{d}t} dt \right|\right| = \left|\left| \int_0^s \nabla L(\theta(t)) dt \right|\right| \leq \int_0^s ||\nabla L(\theta(t))|| dt \tag{13}$$

Using Cauchy-Schwartz,

$$\left(\int_0^s 1 \cdot ||\nabla L(\theta(t))|| dt \right)^2 \leq s \int_0^s ||\nabla L(\theta(t))||^2 dt \tag{14}$$

Thus,

$$||\theta(s) - \theta(0)||^2 \leq s \int_0^s ||\nabla L(\theta(t))||^2 dt < \infty \tag{15}$$

Thus, $||\theta(s)|| < \infty$ for all $s > 0$. □

Theorem A.4. *Let $f : \mathbb{R}^P \to \mathbb{R}^P$ be a piecewise differentiable function (with pieces $E_1, \ldots, E_p$ such that $\cup_{i=1}^p E_i = \mathbb{R}^P$). Let $A \subset \mathbb{R}^P$ such that $\lambda^P(A) = 0$. Then, $\lambda^P(f(A)) = 0$.*

Proof. We know that $f|_{E_i}$ will map zero measure sets to zero measure sets (see Rudin [2006]). Note that $A = A \cap \mathbb{R}^P = A \cap (\cup_{i=1}^p E_i) = \cup_{i=1}^p (A \cap E_i)$. Thus, $\lambda^P(f(A)) \leq \sum_{i=1}^p \lambda^P(f(A \cap E_i)) = 0$. □